

5(a)(i)	any two from: 1 idea of the cells have the, same receptor / specific receptor (for LL-37) ; 2 on cell (surface) membrane / inside cell ; 3 (shape of) receptor is complementary to (shape of), cathelicidin / signalling compound / ligand ; I antigen R active site	2
5(a)(ii)	any two from: 1 thin cells ; A (they are) squamous R 'made of squamous epithelial cells' I flat / flattened R 'they have thin (cell) walls' 2 smooth / AW, (luminal) surface ; 3 cells are wider in region of the nucleus ; A nucleus bulges from cell 4 ref to, endothelial pores / fenestrations / pores between cells / gaps between cells ;	2
5(b)(i)	(TTC to TAC) change from phe to tyr (at position 5) ; I 'tyr' unqualified	1

Question	Answer	Marks
5(b)(ii)	<i>allow two marks for</i> new sequence is leu arg val ile ser ser gly asp leu ;; <i>or allow any two from:</i> I refs to frameshift 1 sequence after, first amino acid / first leu, is different ; R 'after position 2' A 'from position 2 onwards' 2 any example ; e.g. amino acid (at position 2) is arg 3 shortened polypeptide / early chain termination ;	2
5(b)(iii)	<i>any two from:</i> peptide is 3 amino acids in length / shortened polypeptide ; stop codon in position 4 ; (third) amino acid is <u>still</u>, gly(cine) / the same ;	2
5(c)	(genetic code) is the, same / similar, in all organisms AW or <i>idea that each triplet codes for the same amino acid in all living organisms ;</i>	1
5(d)	<i>any two from:</i> 1 (most) amino acids have more than one, triplet / codon ; 2 any correct example(s) from Table 5.2 ; <i>if award this then award mp1 as well</i> 3 genetic code is, degenerate / redundant ; R degenerative / degenerated 4 <i>idea that 64 / 61, possible codons for 20 different amino acids ;</i>	2